Rational Number Project

Initial Fraction Ideas Lesson 9: Overview	Materials
Students continue to explore fraction equivalence using fraction circles to find missing parts of equivalences presented using symbols.	• Fraction Circles for students and teacher • Student Pages A and B for students • Fraction numeral cards for the teacher (cut out)

Teaching Actions

Warm Up

Ty and Shanya both ate part of a Hershey candy bar. Ty ate ½ of his while Shanya ate $\frac{4}{8}$ of hers. Who ate more? Draw a picture to justify your answer.

Large Group Introduction

1. Start the lesson by asking students to take out their fraction circles and find two fractions equal to $\frac{1}{3}$. Ask: What does it mean for two fractions to be equal? From your model, why does $\frac{1}{3}=\frac{2}{6}$?

2. Pose this question: Does $\frac{1}{4}=\frac{4}{8}$?

3. Encourage students to answer this question by either showing with the fraction circles that 1blue = 2 grays or by mentally picturing the fraction circles to conclude that 4/8 = ½ and not ¼.

4. Pose this question: ¾ = how many eighths? Ask how can you record that equivalence with symbols? ($\frac{3}{4}=\frac{6}{8}$)

Small Group/Partner work

5. Students continue to explore fraction equivalences using their fraction circles translating from symbols to manipulatives to symbols. Assign Student Pages A

Comments

1. We want students to think about equivalence as covering up the same amount of the fraction circle. In the next lesson the idea of equivalence will extend to paper folding where students will see that folding the paper into more parts does not change the total amount shaded. Folding only increases the number of parts.

Teaching Actions

and B.

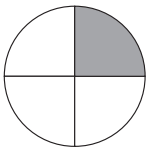
Wrap Up

5. Play the Fraction Fill game.

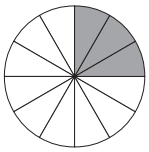
MATERIALS**:** Fraction numeral cards
Fraction Fill Board

DIRECTIONS**:** Teacher randomly selects a numeral card and shows it to students. Students choose to shade that amount on one of the circles. They can only shade 1 representation for that fraction amount.

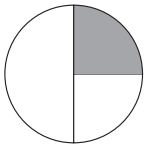
Ex: $\frac{1}{4}$ Student can shade



or



but not both



Student cannot shade the circle divided into two equal parts by adding lines to the circle to divide it into fourths.

Ask students from time to time, what equivalences they used. Record them on the board. Continue showing numeral cards. Students refer to equivalence chart to make selections. The first to shade two complete circles says "Fraction Fill."

Comments

You may want to list some common equivalences on the board as reference. You and the students can construct them together. For example:

$\frac{1}{2} = \frac{2}{4} = \frac{3}{6} = \frac{6}{12}$

Translations

- Written symbols to manipulative to written symbols

Ty and Shanya both ate part of a Hershey candy bar.

Ty ate $\frac{1}{2}$ of his while Shanya ate $\frac{4}{8}$ of hers.

Who ate more? Draw a picture to justify your answer.

Practicing Fraction Equivalence with Our Fraction Circles

Show the Equal Fractions with your Fraction Circles	Record the Equivalent Fractions with Symbols
1 yellow = _____blues	$\frac{1}{2} = \frac{2}{4}$
2 blues = _____reds	
1 brown = _____whites	
4 blues = _____grays	
2 pinks = _____browns	
1 brown = _____pinks	
1 blues = _____reds	

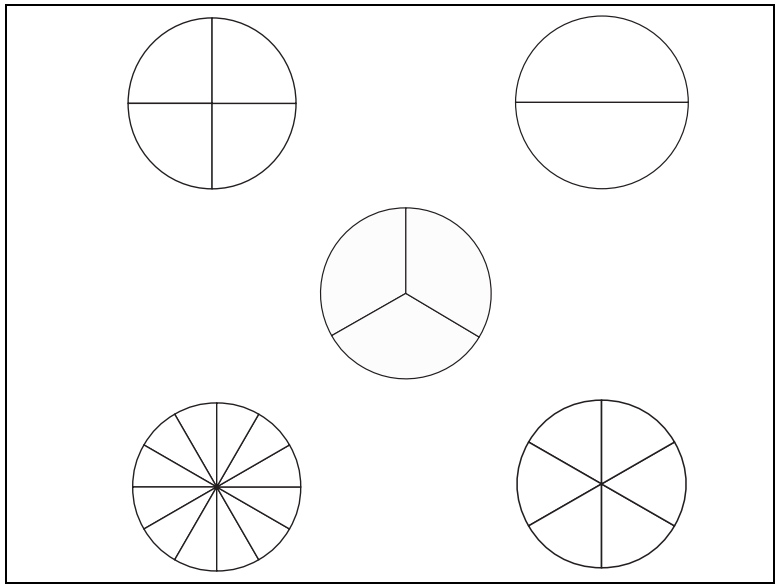
The black circle is the unit or the whole. Use your fraction circles to determine if the following number sentences are **true** or **false**. Circle TRUE if you think the two fractions are equivalent. Circle FALS of you think the two fractions are not equivalent.

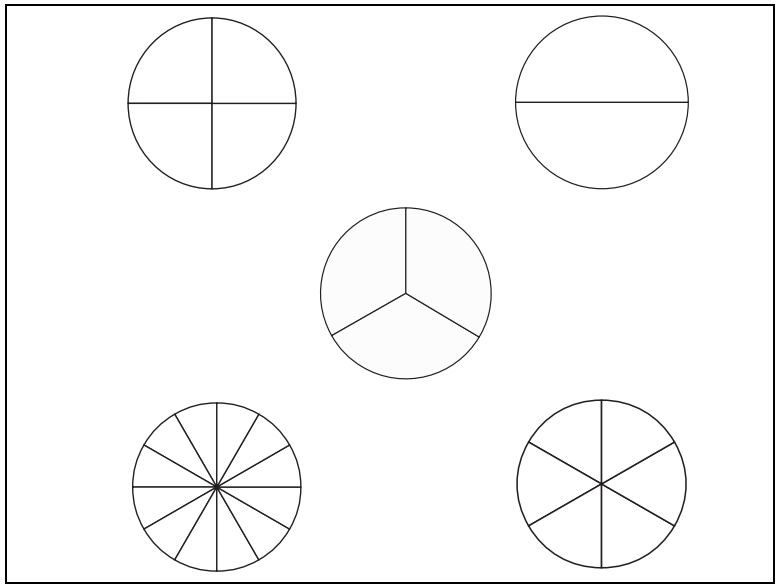
$\frac{1}{2} = \frac{4}{6}$ TRUE or FALSE	$\frac{1}{3} = \frac{2}{6}$ TRUE or FALSE
$\frac{1}{4} = \frac{4}{6}$ TRUE or FALSE	$\frac{3}{12} = \frac{1}{4}$ TRUE or FALSE
$\frac{1}{3} = \frac{4}{12}$ TRUE or FALSE	$\frac{1}{2} = \frac{5}{8}$ TRUE or FALSE

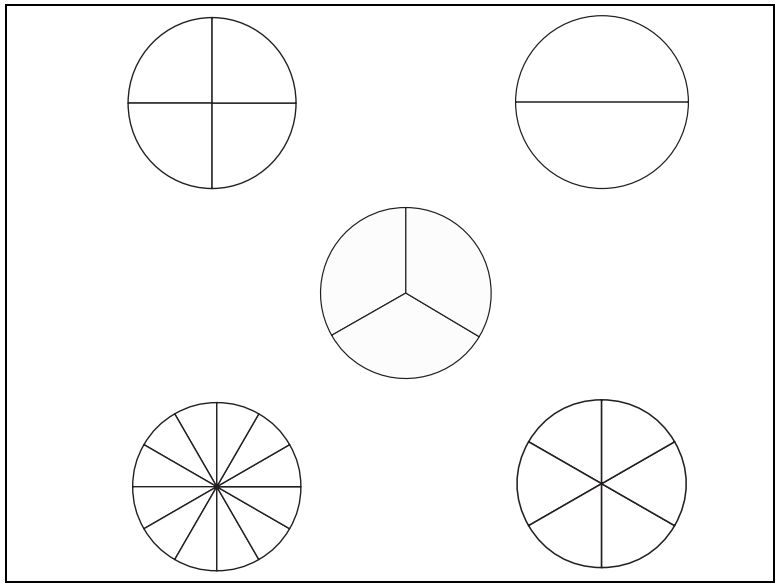
Use your fraction circles to find the missing numerator in each pair to make the number sentences equal.

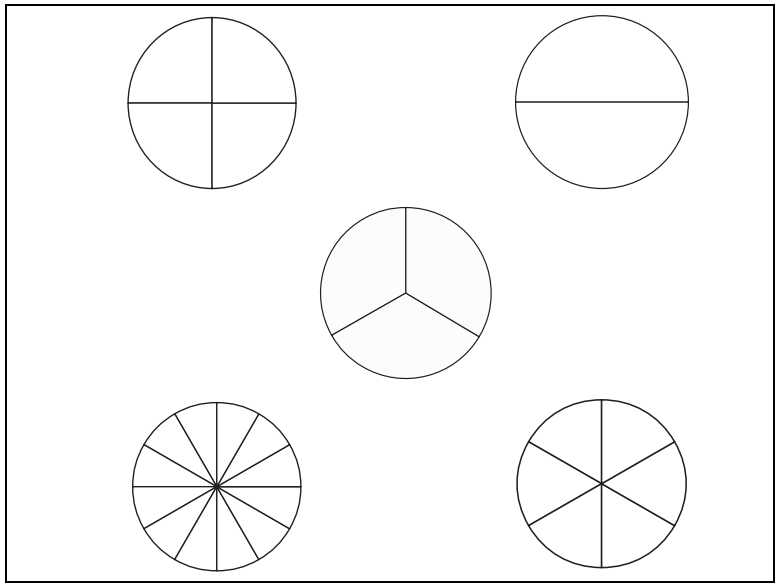
$\frac{1}{2} = \frac{\quad}{6}$	$\frac{1}{3} = \frac{\quad}{9}$
$\frac{2}{2} = \frac{\quad}{6}$	$\frac{3}{4} = \frac{\quad}{12}$

Fraction Fill









$\frac{1}{2}$	$\frac{1}{3}$	$\frac{2}{3}$	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{3}{4}$
$\frac{1}{6}$	$\frac{2}{6}$	$\frac{3}{6}$	$\frac{4}{6}$	$\frac{5}{6}$	
$\frac{1}{12}$	$\frac{2}{12}$	$\frac{3}{12}$	$\frac{4}{12}$	$\frac{5}{12}$	$\frac{1}{12}$
$\frac{7}{12}$	$\frac{8}{12}$	$\frac{9}{12}$	$\frac{10}{12}$	$\frac{11}{12}$	